

E-commerce Sales Data Analysis Report

1. Introduction

This report presents an analysis of E-commerce Sales Data for the years 2024-2025. It covers data loading, cleaning, feature engineering, exploratory data analysis, advanced data validation, aggregation, and data warehousing schema design.

2. Data Overview and Preprocessing

The dataset contains transaction records including order details, customer information, product data, and sales figures. Initial preprocessing included format standardization and missing value audits.

2.1 Raw Data Sample (Split View)

Order ID	Order Date	Customer Name	Region	City	Category	Sub-Category	Product Name	Quantity
10001	2024-10-19 00:00:00	Kashvi Varty	South	Bangalore	Books	Non-Fiction	Non-Fiction Ipsum	2
10002	2025-08-30 00:00:00	Advik Desai	North	Delhi	Groceries	Rice	Rice Nemo	1
10003	2023-11-04 00:00:00	Rhea Kalla	East	Patna	Kitchen	Juicer	Juicer Odio	4
10004	2025-05-23 00:00:00	Anika Sen	East	Kolkata	Groceries	Oil	Oil Doloribus	5
10005	2025-01-19 00:00:00	Akarsh Kaul	West	Pune	Clothing	Kids Wear	Kids Wear Quo	1

Unit Price	Discount	Sales	Profit	Payment Mode	Gross Revenue	Order Year	Order Month	Year_Month	Day_of_Week
36294	5	68958.6	10525.09	Debit Card	72588	2024	10	2024-10	Saturday
42165	20	33732.0	6299.66	Debit Card	42165	2025	8	2025-08	Saturday
64876	20	207603.2	19850.27	Credit Card	259504	2023	11	2023-11	Saturday
37320	15	158610.0	36311.02	UPI	186600	2025	5	2025-05	Friday
50037	10	45033.3	9050.04	Debit Card	50037	2025	1	2025-01	Sunday

2.2 Numerical Summary (Split View)

Metric	Count	Mean	Std
Order ID	5000.0	12500.5	10001.0
Order Date	5000	2024-09-30 18:50:00	2023-10-04 00:00:00
Quantity	5000.0	2.9926	1.0
Unit Price	5000.0	39760.9046	222.0
Discount	5000.0	10.051	0.0
Sales	5000.0	106733.20487	264.1
Profit	5000.0	15941.746981999999	19.12
Gross Revenue	5000.0	118634.2178	278.0

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Order Year	5000.0	2024.2532	2023.0
Order Month	5000.0	6.4798	1.0

Min	25%	50%	75%	Max
11250.75	12500.5	13750.25	15000.0	1443.5200033252052
2024-04-05 00:0...	2024-09-29 00:0...	2025-03-30 00:0...	2025-10-03 00:0...	NaT
2.0	3.0	4.0	5.0	1.4131329126893282
20312.25	39459.5	59721.75	79998.0	22831.78394601545
5.0	10.0	15.0	20.0	7.084662129381769
39766.5375	83080.325	156968.5875	398485.0	85108.2082018013
4892.295	11108.525000000001	22467.987500000003	89688.44	14897.68491575783
44950.5	94378.0	173078.25	399400.0	93792.62489266654
2024.0	2024.0	2025.0	2025.0	0.6554202278958169
4.0	6.0	9.0	12.0	3.4433070198061784

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3. Key Visualizations Dashboard

The dashboard below summarizes monthly revenue trajectories, top revenue categories, regional order counts, and daily shopping volumes.

